METHOD REFERENCE · TCPS-PA v3.0

Event-Centric · Dynamic Timing Contract · Bidirectional Diagnosis · Physical Budget Loss

CARLA 0.9.15 + Apollo 10.0.0 + Bridge

run

Observed / Requirement / Model / Retrospective

2026-08-13

run

“ run ” “ ” run

$$U_{c,k}=(e_k, P_{c,k})$$

\(e_k\) \(k\) Safety-Relevant Event \(P_{c,k}\) \(c\) source-to-physical-effect Cause-Effect Chain run

L1 L6

□□□	□□□	□□□□□□□
L1 Temporal Stressor / Root-Cause		
L2 Local Temporal Degradation		task edge sample-and-hold
L3 Cause-Effect Timing		Reaction Age Fresh Gap
L4 Dynamic Temporal Contract		miss
L5 Physical Budget Loss		
L6 Physical Safety State		

```
L4/L6 seed -> L3 lineage -> L2 local symptom -> L1 root-cause hypotheses
L4 contract state -> L5 physical budget loss -> L6 physical outcome

L1 verified intervention -> L2 -> L3 -> L4 -> L5 -> L6 -> attribution
```

—

1. chain task/edge/actuation segment
2. execution queue activation/phase transport staleness reuse gap overwrite/drop actuation / /
3. observed miss guarantee loss observed miss
4.
5. deadline-excess distance debt
6. SUT

1.

1.1 “ ”

“ ”

-
-
- miss
-
- **SUT** Apollo/Bridge run “ ” “ ”

latency gap

1.2

—

7. **Observed timing correctness**
8. **Analytical/formal timing guarantee**
9. **Single-run empirical headroom** run

WCET/WCRT / actuation bound run max P99 formal guarantee

> run

run

1.3

□□□	□□	□□□□□
data/observed		
requirement/qualified		
model/predicted		
retrospective/reconstructed	run	

data/observed UNAVAILABLE

1.4

Apollo Bridge Guardian Bridge Control

```
... -> Planning -> Control -> Bridge -> CARLA vehicle dynamics -> physical response
```

Guardian Control

---

2. Run Safety-Relevant Event Cause-Effect Chain

2.1 Run

run sensor source effect run E2E latency

-
- source-to-sink path
-
-
-

```
run_id + event_id + chain_id + chain_instance_id
```

2.2 Safety-Relevant Event

Safety-Relevant Event “ ”

```
E_k = {
  event_id, event_type, target_id, scenario_geometry,
  cause_predicate, t_c, t_c_interval, clock_domain,
  state_at_cause, discovery_method, prospective_or_retrospective,
  source_evidence_ids, uncertainty, limitations
}
```

- /
-
- /
-
- Planning STOP/ /
-

\(t_c\) cause_predicate \
([t_c^L,t_c^U]\)

2.3 deadline

run discovery RETROSPECTIVE_EVENT_DISCOVERY

deadline cut-off

```
event discovery may inspect the future
deadline construction may not use post-cause outcome
```

run \(t_{phys}\) run

2.4 Cause-Effect Chain

DAG source-to-sink path source latency

```
LiDAR -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
Radar -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
Localization -> Planning -> Control -> Bridge -> physical effect
Chassis -> Control -> Bridge -> physical effect
```

ECRTS 2023 source/sink DAG source-to-sink cause-effect path Yi deadline sensor-to-actuator path task E2E deadline [ECRTS 2023](#) [Yi et al. 2021](#)

2.5 effect

“ ” effect effect predicate

```
Psi_effect(event_type, target_id, vehicle_state, command_state) = TRUE/FALSE
```

effect

- t_cmd Control
- t_apply Bridge /
- t_phys effect predicate
- effect_hold_rule

effect E2E physical Reaction Time

3.

3.1

□□□	□□□□□□□	□□□□□□□□□
Bridge/SCB	delay	deadline miss

□□□	□□□□□□□□	□□□□□□□□□□
Apollo	fallback solver	
Trace events	read/start/end/pub / monotonic	/
Trace message context	trace_id parent input/output seq source data timestamp	target outcome
Parsed Cyber record	channel /record receipt header/source timestamp payload reuse	read/start/finish
Localization/Chassis		CARLA sim time
Collision event	actor impulse	
Actor history/ground truth	CARLA frame/sim	Apollo

Apollo Record Cyber RT channel / record timeline
[Apollo 10 Cyber RT Developer Tools]

3.2

timestamp clock_domain host timestamp_type unit resolution alignment_method error_bound

- wall_epoch epoch
- monotonic monotonic trace
- record_time / record
- header_timestamp header
- measurement/source_time
- CARLA_sim_time/frame
- Bridge_apply_wall Bridge

10.
11.
12. offset/drift/anchor
13. age
14.

P_CLOCK “ ” t_c t_{phys} Orin

3.3 phase

“ timestamp ” phase “producer consumer CARLA tick / ” phase quantization phase offset

- clock_alignment_audit.csv
- phase_audit.csv

phase latency PHASE_EFFECT_HYPOTHESIS phase

3.4

```
DIRECT_OBSERVED
OBSERVED_DERIVED
TRACE_LINEAGE
WEAK_TEMPORAL_ALIGNMENT
INDEPENDENT_REQUIREMENT
INDEPENDENT_CALIBRATED_MODEL
VALIDATED_MODEL
UNVALIDATED_MODEL
RETROSPECTIVE_RECONSTRUCTION
OUTCOME_TRUNCATED
CONFLICTED_EVIDENCE
MISSING
```

CLOCK_TAINT TARGET_TAINT MODEL_TAINT RETRO_TAINT OUTCOME_TRUNCATED

3.5

- P_TARGET sensor/Fusion/Prediction/Planning/Control/ actor
- P_FUNC Perception Prediction Planning trajectory Control trajectory Bridge

P_FUNC=QUALIFIED_PASS deadline “ ” Planning STOP trajectory infeasible Control

4. L3 Cause-Effect Timing

run L3/L4 L6 seed L1

4.1

trace record Localization Bridge

```
event_time.csv
{
  event_uid, run_id, event_id, chain_id, module, stage,
  event_kind, read_or_write, clock_domain, host,
  raw_time, aligned_wall_time, alignment_error,
  input_seq, output_seq, trace_id, parent_trace_id,
  data_source_time, target_id, payload_digest,
  source_file, source_locator, evidence_class, quality_flags
}
```

event_kind physical_cause sensor_sample receive/read ready start end publish/write command bridge_apply physical_effect outcome

4.2

Grade	□□□□	□□□□□□□
A	trace_id parent/provenance input/output seq	cause-effect propagation
B	source/header timestamp + downstream mapping	propagation
C		/
D	nearest-time heuristic	
UNKNOWN		

P99 stage decomposition causal chain instance

4.3 Forward causal tracing

source sample “ source ” job

```
source sample -> earliest eligible downstream read -> ... -> first qualifying command -> physical effect
```

- source data sink
- consumer
-
- /

- Control Bridge

4.4 Backward provenance tracing

effect Control Planning

```
physical effect <- applied command <- Control input trajectory
        <- Planning inputs <- Prediction/Fusion source samples
```

“ ” Data Age reuse/staleness Forward backward

- forward sample effect
- observed effect sample
- overwrite skip reuse

4.5 ECRTS partition

ECRTS 2023 p-partitioned job chain backward chain forward chain warm-up MRT MDA partition point

```
at partition p:
backward provenance from p to source
+ forward propagation from p to physical effect
```

-
- read/write job warm-up/valid-chain job-chain
- PARTITION_DIAGNOSTIC_VIEW MRT/MDA

4.6 Multi-input Age Vector

Planning/Control Age

$$A_{c,k}(t_e)=[A_{perception},A_{prediction},A_{localization},A_{chassis},A_{map},\ldots]$$

$$A_j=t_{effect/read}-t_{source,j}$$

source semantics chain target relevance clock error $A_{critical}$ vector critical source

4.7 Fresh Causal Update Gap

output gap effect q_n

```
qualifying(q_n) iff
  target/event semantics match
  AND source timestamp strictly advances
```

```
AND lineage reaches the selected sink/effect
AND output is not mere reuse of the same source state
```

$$G^{fresh}_n=t(q_{n+1})-t(q_n)$$

- G_publish
- G_source_advance source timestamp
- G_fresh_effect sink/effect

“Control 100 Hz ” “ 10 ms ” Control Planning trajectory

---

5. L2

5.1 task/edge

producer $(i-1)$ consumer (i)

```
t_write_prev  producer
t_receive          consumer/record
t_ready          job
t_start      job
t_end        job
t_write      consumer
```

$$L_i^{transport}=t_{receive}-t_{write,prev}$$

$$L_i^{queue}=t_{ready}-t_{receive},\ L_i^{activation}=t_{start}-t_{ready}$$

$$L_i^{exec}=t_{end}-t_{start},\ L_i^{publish}=t_{write}-t_{end}$$

ready start-receive queue phase unresolved_wait record
record

5.2

□□	□□□	□□□□□□
Sampling delay	t_s-t_c	sensor tick phase
Transport delay	receive - producer write	SHM/RTPS bridge
Queue/backlog	ready - receive /	CPU/GPU contention callback backlog
Activation/phase wait	start - ready	timer/callback phase

□□	□□□	□□□□□□□
Execution tail	end - start	fallback
Publish delay	write - end	writer/backpressure
Staleness	consumer time - source time	producer
Reuse	sink source/trajectory	sample-and-hold gap
Overwrite/skip	source update downstream successor	buffer policy
Drop/reorder	seq /	transport/queue policy
Actuation delay	physical effect - applied command	Bridge

5.3 R/A/G

T_deg=[R,A,G]

- R reaction/response-time behavior
- A data freshness/age
- G update continuity/gap

run count P50 P90 P95 P99 MAX IQR/MAD run run formal bound

5.4 run “ ”

L2 “ ” reference run

- NOMINAL_PERIOD
- ENGINEERING_REQUIREMENT /edge timing requirement
- PRE_FAULT_WITHIN_RUN run
- DECLARED_EXTERNAL_REFERENCE
- LOCAL_STATIONARY_REFERENCE run

context covariates defeater

reference

> run 412 ms Planning

> Planning continuity degraded

5.5

Prediction -> Planning 85 ms edge/segment phase queue scheduler interference ready/start / L2

```
observed symptom
candidate mechanisms
supporting evidence
challenging evidence
missing discriminator
```

6. L1 Temporal Stressor

6.1

Discovery mode L1 L4 miss L6 loss L3/L2 anomaly

Validation mode L1 " delay" delay

6.2 Temporal Fault/Stress Signature

```
F_T = {
  fault_or_stressor_type, location, onset, end, duration,
  requested_magnitude, actual_magnitude, distribution,
  persistent_or_transient, one_shot_or_repeated,
  affected_channels/messages, queue_policy,
  drop_status, reorder_status, resource_context,
  direct_evidence, confidence
}
```

- sampling/trigger delay
- execution/WCET tail
- CPU/GPU/memory/lock contention
- scheduler/priority/blocking
- queue/backlog/backpressure
- network/transport/Bridge delay
- period/phase/mode-transition delay
- staleness/reuse/overwrite/drop/reorder
- actuation/control-build-up delay

- clock/timestamp artifact
- functional or physical alternative

6.3

“ ” applied row Bridge/SCB Bridge/Control

Apollo

fault onset t_c $[t_f,t_c]$ pre-hazard state divergence

/

---

7. L4 Per-Event / Per-Chain Dynamic Temporal Contract

L4

7.1 deadline

x_c ρ / θ H

$$\Phi(x_c,\rho;\theta)=\min_{t\in[0,H]} h(x_{ego}(t),x_{target}(t),geometry,policy)$$

$\Phi\ge0$ dynamics safety buffer deadline

$$\tau_{R,c,k}^{req}=\sup\{\rho\ge0\mid \in f_{\theta\in\Theta_m}\Phi(x_c,\rho;\theta)\ge0\}$$

Θ_m context/ ODD deadline x_c

7.2 /

RSS-like

$$d_{req}(\rho)=v_e\rho+\frac{1}{2}a_{resp}\rho^2+\frac{(v_e+a_{resp}\rho)^2}{2b_e}-\frac{v_f^2}{2b_f}+d_{safe}$$

$d_{req}(\rho)\le d_{clear}$ ρ $v_f=0$ $\rho=0$

```
tau_req = 0
state = ALREADY_OUTSIDE_DECLARED_ENVELOPE
```

delay

Koopman Osyk Weast longitudinal model response period braking trajectory closest approach Koopman et al. 2019

reachability/trajectory model deadline
NOT_QUALIFIED_PRIMARY

7.3 Deadline

deadline observed miss

- state input t_c cut-off
- target identity geometry target-motion assumption
- / dynamics braking envelope
- calibration runs evaluation run
- friction grade curvature actuation build-up uncertainty bounds
- run post-t_c t_{phys}
- tau_req_low/center/high provenance

UNVALIDATED_MODEL RETROSPECTIVE_RECONSTRUCTION
NOT_QUALIFIED_PRIMARY

7.4 ODD / context uncertainty

μ context region m

$$\tau_{robust}(m)=\in f_{x\in X_m}\tau(x)$$

“ region ” ODD “region ” “ region ”
[Koopman et al. 2019]

7.5

Reaction Freshness Fresh Update Continuity

$$S_{obs}=[\tau_R^{req}-T_R^{obs}; \tau_A^{req}-A_{critical}^{obs}; \tau_G^{req}-G_{fresh}^{obs}]$$

τ_A^{req} τ_G^{req} τ_R^{req}

- Freshness contract “ a / ”
- Fresh-update contract “ g / reachable states ”
- A/G freshness/gap requirement reaction deadline

ECRTS warm-up MRT MDA job-chain observed Reaction Age A/G [ECRTS 2023]

7.6 observed contract verdict

$[T_L,T_U]$ deadline $[\tau_L,\tau_U]$

□□	□□
\(T_U\le\tau_L\)	CLEARLY_WITHIN_CONTRACT
\(T_L>\tau_U\)	CLEARLY_MISSED
	BOUNDARY_UNCERTAIN
deadline	RETROSPECTIVE_ONLY
	MODEL_SUPPORTED_ONLY
/clock/target	NOT_TESTABLE

observed slack

7.7 Observed correctness Timing Guarantee

$$S\_obs=\tau\_R^req-T\_R^obs$$

$$S\_guar=\tau\_R,low^req-R\_path^UB$$

\(R_{path}^{UB}\) / / WCET/WCRT release/period priority/scheduler blocking queue/transport multi-rate overwrite semantics phase actuation bound

Yi \(\sum_i(p_i+r_i)\) \(2\sum_i p_i\) CPU EDF WCET Apollo budget Apollo trace period formal guarantee Yi et al. 2021

\(S_{obs}\ge0\) \(S_{guar}<0\)

\(R^{UB}\) S_guar=NOT_ESTABLISHED

> single_run_empirical_headroom = tau_req - max observed comparable instance

guarantee

7.8 Dynamic deadline shrink mode transition

Yi deadline deadline job sensor data E2E deadline Yi et al. 2021

hazard window contract trajectory

$$\dot S(t)=\dot\tau^req(t)-\dot T^risk(t)$$

$S>0,\dot S<0$

$$T_{exhaust}^{local}=\frac{S}{-\dot S}$$

deadline region/mode change job trajectory contract

7.9 Slack Waterfall Guarantee-Loss Point

chain i

$$L_{prefix,i}^{obs}=t_i-t_c$$

$$B_i^{obs}=\tau_R^{req}-L_{prefix,i}^{obs}$$

i effect suffix bound $R_{i\rightarrow phys}^{UB}$

$$B_i^{guar}=\tau_{R,low}^{req}-L_{prefix,i}^{obs,U}-R_{i\rightarrow phys}^{UB}$$

sampling transport queue/unresolved wait activation execution publish consumer wait command-to-apply actuation

15. FIRST_CONDITIONAL_GUARANTEE_LOSS $B_i^{guar}<0$
16. FIRST_IRREVERSIBLE_OBSERVED_MISS effect τ_U miss
17. OBSERVED_EFFECT_MISS t_{phys} miss

FIRST_CONDITIONAL_GUARANTEE_LOSS " + "

segment budget/reference

suffix bound observed slack waterfall guarantee-loss point

---

8. seed

8.1 seed

18. L4 contract miss
19. L6 L4
20. L3 physical Reaction/Age/Fresh Gap
21. L2 reference

22. reference case-level anomaly

seed C1

8.2 Backward diagnosis

seed

23. **seed** chain outcome
24. **lineage segment** effect/command source budget
25. sampling execution queue activation/phase transport staleness/reuse overwrite/ drop mode transition Bridge/actuation
26. clock artifact target mismatch functional failure initial unsafe state braking/road/geometry outcome conflict
27. onset seed
28. supporting challenging/contradicting missing evidence
29. equivalence class
30. **discriminator** trace /
31. /

8.3

```
SUPPORTED_CANDIDATE
CONSISTENT_BUT_UNRESOLVED
CHALLENGED
REFUTED
NOT_TESTABLE
NOT_APPLICABLE
```

□□	□□
DETECTED	/
LOCALIZED_TO_SEGMENT	
MECHANISM_SUPPORTED	
ISOLATED_CANDIDATE	
INTRINSIC_DEFECT_SUPPORTED	/ /

run “ ” LOCALIZED_TO_SEGMENT ISOLATED_CANDIDATE

8.4 Forward validation

```
verified stressor
-> matching local primitive changes
-> causally matched chain timing effect
-> qualified contract state
-> physical budget loss
-> outcome
```

backward candidate

9. L5 Physical Safety Budget Loss

9.1 Canonical observed response distance

cause \(t_c\) effect \(t_{phys}\)

D_response,wall^data/observed= ∈t_t_c^{t_phys}v(t) dt_wall

```
D_response_wall_integral_data_observed_m
```

D_delay_wall_integral_data_observed_m CARLA frame/sim time Localization realtime factor

9.2 Deadline-excess distance debt

deadline endpoint

t_d=t_c+τ_R^req

miss

D_debt^requirement\ constrained= max≤ft(0,∈t_t_d^{t_phys}v(t) dt_wall)

" " ancestry " requirement + observed velocity path" REQUIREMENT_CONSTRAINED_DERIVED

- model deadline -> D_debt_model_predicted_m
- retrospective deadline -> D_debt_retro_diagnostic_m
- deadline -> D_debt

9.3

M_0^data/observed=D_1-D_response^obs-D_brake^obs

$$M\_safe^{data/observed}=M\_0^{obs}-D\_safe$$

\(D_1\) \(D_{response}\) \(D_{brake}\) fault onset \(t_c\)

- Total Closed-loop Effect \(t_f\) outcome
- Post-cause Response Effect \(t_c\)

response distance deadline-excess debt

9.4

/

$$M\_phys^{data/observed}= \min_{t\in[t\_c,t\_o]} h\_observed(x\_e(t),x\_o(t),geometry)$$

CARLA ground truth/actor history outcome Apollo “ ”

$$M\_phys^{model/predicted}(\rho)= \min_t h\_model(x\_e(t;\rho),x\_o(t),\Theta\_m)$$

slack margin margin margin

9.5 “ ”

run

32. D_response_wall_integral_data_observed_m
33. D_debt_requirement_constrained_m deadline
34. M_phys_data_observed_m / outcome
35. Delta_M_phys_model_predicted_m \(\rho_{ref}\)

$$\Delta M\_phys^{model}=M\_phys^{model}(\rho\_ref)- M\_phys^{model}(T\_R^{obs})$$

4 “ timing X ”

9.6 Collision right-censoring

run

- collision actor time/frame impact speed/impulse
- observed response/braking segment
- actual clearance/contact state

“ observed braking distance full-stop margin”

UNAVAILABLE_OUTCOME_TRUNCATED observed

10. L6 Physical Safety State

L6

- minimum/final clearance
- physical/contact/engineering margin
- minimum speed strict stop near stop
- impact speed impulse collision actor
- truncated braking time/distance
- safety envelope violation duration/area

outcome

```
SAFE
SAFE_LOW_MARGIN
CRITICAL_MARGIN
MINIMUM_DISTANCE_VIOLATION
EMERGENCY_MANEUVER
NEAR_MISS
CONTACT
COLLISION
HIGH_SEVERITY_COLLISION
UNCERTAIN
```

CONTACT/COLLISION low/critical/near/high-severity provenance

/

11. run

Step 0

- run inventory collect window trace collision actor history record Bridge/SCB
- run analysis directory
- run record run record

Step 1

- profile parse error channel
- timestamp dictionary clock alignment phase audit
- NOT_TESTABLE “ ”

Step 2

- conflict predicate margin local minimum contact/collision candidate events
- candidate t_c target geometry effect predicate discovery provenance
- / candidate IDs

Step 3

- P_TARGET P_FUNC
- fallback/infeasibility
- Guardian Bridge

Step 4 paths

- DAG/config/trace/record channel source-to-effect paths
- path forward causal tracing backward provenance
- lineage grade

Step 5 L3 observed chain metrics

- $T_{sampling}=t_s-t_c$
- $T_{software}=t_{cmd}-t_s$
- $T_{actuation}=t_{phys}-t_{cmd}$ command-to-apply + apply-to-physical
- $T_R=t_{phys}-t_c$
- Age Vector publish/source-advance/fresh-effect gaps
- uncertainty

Step 6 L2

- chain instance transport/wait/activation/exec/publish/reuse
- -run reference
- reference
- local symptom

Step 7 L4 contract

- safety model
- cut-off state dynamics/uncertainty
- τ_R τ_A τ_G low/center/high
- anti-leakage domain qualification

- model/retrospective observed contract verdict

Step 8 observed correctness guarantee

- uncertainty-aware contract verdict
- suffix/WCRT bounds slack waterfall guarantee-loss points
- formal bound observed waterfall single-run empirical headroom
- shrinking contract mode-transition/old-job/old-trajectory

Step 9 Backward diagnosis L1

- miss/loss seed segment
- hypothesis ledger equivalence classes defeaters discriminators
-

Step 10 L5/L6

- observed wall-clock velocity $D_{response}$
- deadline D_{debt}
- margin/outcome
- observed
- collision run right-censoring

Step 11

- clock phase target function initial state braking geometry freshness outcome conflict defeaters
- Evidence Ledger -> Claim Ledger -> Defeater Ledger
-

12.

12.1

- Perception target / tracking continuity
- Prediction target/trajectory
- Planning STOP/decision/fallback/infeasibility/latency debug
- Control command/latency

- Localization speed/acceleration/pose
- Bridge/SCB requested/applied delay
- collision and actor history

timestamp

12.2 Trace

e2e_trace_v3

```
events: event_id, trace_id, module, phase, mono_ns, pid, tid
message_context: edge, channel, input_seq, output_seq,
               data_ts_ns, primary_parent_trace_id,
               parent_trace_count, trace_valid
```

- Grade A/B lineage
- proc/runonce/substage timing
- source timestamp
- parent/fusion lineage
- overwrite/skip

monotonic trace trace anchor uncertainty

12.3 Parsed Cyber record

record

- record_message_index channel coverage/rate/gap
- Localization/Chassis state
- Planning header task latency decision trajectory
- Control command latency vehicle state Planning reuse/age
- Perception/Prediction obstacles source headers
- sensor timeline
- Planning-Control alignment sensor-to-Control message diagnostics

- channel protobuf
- record_time / task start
- sensor_to_control_reaction_time source-to-Control t_c t_{phys} physical Reaction Time
- obstacle table obstacle row gap channel timeline

- Planning total_time_ms task row frame statistic planning_seq
  run record
- run record
- record-only Age/reuse/message payload
- trace/log
- L3 lineage L2 mechanism

12.4 Collision

- collision event/JSONL outcome
- actor history realtime factor
- Localization wall-time velocity
- CARLA frame/sim time e2e_sim_frames realtime_factor

13.

```
analysis/<run_id>/
├── report/
│   └── single_run_realtime_diagnosis.md
├── tables/
│   ├── input_inventory.csv
│   ├── event_catalog.csv
│   ├── event_timeline.csv
│   ├── chain_catalog.csv
│   ├── chain_instances.csv
│   ├── partition_lineage.csv
│   ├── local_timing_primitives.csv
│   ├── age_vector.csv
│   ├── fresh_update_gaps.csv
│   ├── temporal_fault_signature.csv
│   ├── clock_alignment_audit.csv
│   ├── phase_audit.csv
│   ├── target_identity_audit.csv
│   ├── functional_correctness_audit.csv
│   ├── requirement_registry.csv
│   ├── dynamic_contract_construction.csv
│   ├── contract_verdicts.csv
│   ├── slack_waterfall.csv
│   ├── guarantee_loss_points.csv
│   ├── velocity_trajectory_observed.csv
│   ├── physical_budget_observed.csv
│   ├── physical_budget_model_predicted.csv
│   ├── physical_outcomes.csv
│   ├── diagnosis_hypothesis_ledger.csv
│   ├── diagnosis_edges.csv
│   ├── evidence_ledger.csv
│   ├── claim_ledger.csv
│   ├── defeater_ledger.csv
│   └── exclusions_and_missing.csv
```

```
└── validation/
    ├── data_quality_audit.md
    ├── anti_leakage_audit.md
    ├── arithmetic_recomputation.md
    ├── claim_audit.md
    └── validation.json
```

13.1

```
Event / Target / Geometry
Cause predicate and t_c interval
Selected critical chain(s) and lineage grade

Observed chain timing
  T_sampling / T_software / T_command_to_apply / T_actuation / T_R
  Age Vector / A_critical basis
  G_publish / G_source_advance / G_fresh_effect

Dynamic contracts
  tau_R low/center/high + qualification
  tau_A and tau_G if independently available
  observed verdict and S_obs
  formal/analytical guarantee status and S_guar
  empirical headroom, explicitly non-guarantee

Slack waterfall
  per-segment observed budget use
  first conditional guarantee-loss point
  first irreversible observed miss
  dominant over-budget segment relative to declared reference

Diagnosis
  localized segment
  mechanism candidates and evidence
  observational equivalence class
  competing functional/clock/phase/physical explanations
  diagnosis strength and missing discriminator

Physical propagation
  D_response_wall_integral_data_observed_m
  D_debt_requirement_constrained_m
  M_phys_data_observed_m
  Delta_M_phys_model_predicted_m
  right-censoring status

Outcome and allowed conclusion
```

---

14. Claim–Evidence

I-M-E-C-O-N

- I / Inputs
- M / Metrics
- E / Evidence /
- C / Criterion

- O / Output verdict confidence ceiling
- N / Next

14.1

ID	□□
C1	stressor/
C2	reference
C3	/ effect
C4	observed timing
C5	timing violation
C6	/
C7	

C7

14.2

Level	□□□□□
0	/
1	reference
2	cause-effect propagation
3	requirement observed contract miss guarantee loss
4	timing
5	timing-dominated safety-failure candidate
6	SUT-intrinsic temporal defect

/

14.3 defeaters

> clock alignment uncertainty
> phase/tick quantization
> target mismatch

```
functional failure or fallback
initial unsafe state
pre-cause treatment-induced state divergence
braking capability / friction / grade / curvature mismatch
geometry or actor association conflict
record/trace window incomplete
outcome truncation
deadline leakage or model-domain mismatch
freshness/reuse alternative
external injected fault versus intrinsic Apollo defect
```

defeater OPEN/UNKNOWN UNCERTAIN/PARTIAL_PASS

15.

15.1

```
CHAIN_IDENTIFIED
SEGMENT_LOCALIZED
TASK_OR_EDGE_LOCALIZED
MECHANISM_SUPPORTED
ROOT_CAUSE_ISOLATED
NOT_LOCALIZABLE
```

“ task” /reference over-consumption

15.2

```
SAMPLING
TRANSPORT
QUEUE_BACKLOG
ACTIVATION_PHASE
EXECUTION
PUBLISH_BACKPRESSURE
STALE_DATA
REUSE_SAMPLE_HOLD
OVERWRITE_SKIP
DROP_REORDER
MODE_TRANSITION
BRIDGE_DELAY
ACTUATION
CLOCK_ARTIFACT
FUNCTIONAL_ALTERNATIVE
PHYSICAL_ALTERNATIVE
UNRESOLVED_MIXTURE
```

15.3

- observed_contract_miss_time

- first_conditional_guarantee_loss_boundary/time
- first_irreversible_observed_miss_time
- contract_shrink_crossing_time
- formal bound GUARANTEE_NOT_ASSESSABLE

15.4

```
supported mechanism
+ causal segment
+ consumed budget relative to reference
+ evidence IDs
+ counter-evidence
+ unresolved equivalent candidates
+ missing discriminator
```

15.5

provenance

```
L_physical= (D_response^obs,D_debt^req,M_phys^obs,Δ M_phys^model)
```

16.

36. collision -> deadline miss
37. deadline miss -> collision fallback
38. output exists -> function correct
39. record_time difference -> execution time record task start/end
40. Control publishes at 100 Hz -> fresh control every 10 ms Planning
41. MRT maximum = MDA maximum -> every RT equals every Age warm-up/chain
42. p-partition point -> root cause proven /
43. single-run P99/max -> formal guarantee run
44. first guarantee-loss point -> fault origin
45. model deadline -> observed deadline miss
46. same-run stopping distance -> independent deadline
47. CARLA sim frame displacement -> D_response

48. collision-truncated trajectory -> full observed stopping distance
49. Bridge injection -> Apollo intrinsic defect
50. D_response + D_debt both subtracted debt response distance

17. run

□□	□□□□	□□□□□□□□
event	ground truth/target/ predicate	EVENT_UNCERTAIN
chain	trace/provenance/headers	ASSOCIATION_ONLY / NOT_TESTABLE
	read/start/end/write	UNRESOLVED_SEGMENT
miss	observed physical RT + qualified deadline	MODEL_ONLY / RETROSPECTIVE_ONLY / NOT_TESTABLE
	qualified path/suffix upper bounds	GUARANTEE_NOT_ESTABLISHED
	+ alternatives	CANDIDATE_SET_ONLY
	wall velocity + endpoints + qualified deadline	RESPONSE_DISTANCE_ONLY / UNAVAILABLE
	collision/geometry/complete trajectory	OUTCOME_UNCERTAIN
Apollo		NOT_ESTABLISHED_FROM_SINGLE_RUN

NOT_TESTABLE run

18.

- event predicate \(t_c\)
- target
- chain causal instance
- forward backward lineage warm-up/overwrite

- Reaction Age Gap
- software command physical effect
- record/header/source/trace/wall/sim time
- clock error slack verdict

Contract

- deadline
- cut-off run outcome
- geometry/model/domain
- low/center/high
- observed correctness formal guarantee empirical headroom

- guarantee-loss point root cause
- overrun reference/budget
- phase/queue/execution discriminator
- backward hypothesis forward proof

- $D_{response}$
- D_{debt} deadline
- moving target closest approach
- observed model
- collision truncation

- defeaters
- claim level
-

19.

19.1 ECRTS 2023

cause-effect chain read/write job immediate forward/backward chain MRT/MDA partition warm-up path

MRT=MDA Reaction=Age partition theorem root-cause localization theorem job read/write

19.2 Yi et al. 2021

sensor arrival E2E deadline source-to-sink path constraint task schedulability E2E guarantee shrinking deadline/mode-transition

CPU EDF WCET \(2\sum p_i\) Apollo/Cyber RT bound — deadline

19.3 Koopman, Osyk & Weast 2019

context-dependent safety envelope response time braking trajectory closest approach friction/slope/curvature/vehicle/environment variability ODD

RSS-like ODD /

20.

—

{ (e_k,P_c,k) ⟶ ≤ft(T_R^obs, A^obs,G_fresh^obs) ∥ ≤ft(τ_R^req,τ_A^req,τ_G^req) }

{ S_obs=τ^req- T^obs}

{S_guar=τ_R,low^req-R_path^UB}

chain observed/guarantee slack waterfall miss L2 primitives L1

{ D_response^obs, D_debt^req, M_phys^obs, Δ M_phys^model, Outcome^obs }

> 300 ms

> \(e_k\) \(P_{c,k}\) observed Reaction/Freshness/Continuity
> requirement-
> constrained distance debt

51. M. Günzel, H. Teper, K.-H. Chen, G. von der Brüggen, J.-J. Chen, "On the Equivalence of Maximum Reaction Time and Maximum Data Age for Cause-Effect Chains," ECRTS 2023. https://doi.org/10.4230/LIPIcs.ECRTS.2023.10
52. S. Yi, T.-W. Kim, J.-C. Kim, N. Dutt, "Energy-Efficient Adaptive System Reconfiguration for Dynamic Deadlines in Autonomous Driving," 2021. https://arxiv.org/abs/2106.04508
53. P. Koopman, B. Osyk, J. Weast, "Autonomous Vehicles Meet the Physical World: RSS, Variability, Uncertainty, and Proving Safety," SAFECOMP expanded preprint, 2019. https://arxiv.org/abs/1911.01207
54. Apollo 10.0, "Apollo Cyber RT Developer Tools." https://apollo.baidu.com/docs/apollo/10.x/md_cyber_2docs_2cyber__developer__tools.html
55. S. Shalev-Shwartz, S. Shammah, A. Shashua, "On a Formal Model of Safe and Scalable Self-driving Cars." https://arxiv.org/abs/1708.06374